

Machine Learning

9. *Regularization*

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Today, we will learn...

- Regularization in ML
 - What is 'regularization'?
 - Why do we need it?



These slides are heavily based on Prof. Kim at Korea University

<https://www.youtube.com/watch?v=pJCcGK5omhE&list=PLpIPLT0Pf7IoTxTCi2MEQ94MZnHaxrP0j&index=26>

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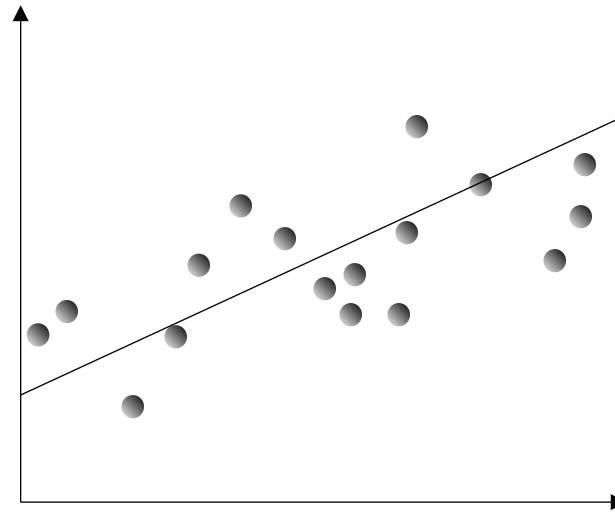
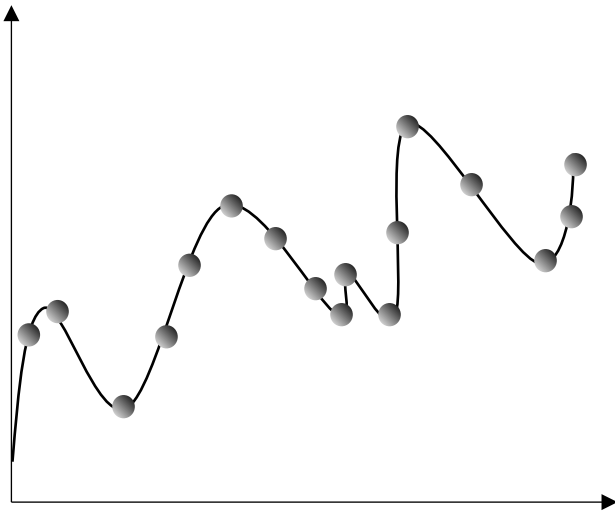
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- ***Good Model?***
- Regularization Concept
- Ridge Regression (L_2 -Regularization)
- Lasso Regression (L_1 -Regularization)

What is a Good Model?

Explanatory Model vs. Predictive Model



Good Explanatory Model

- Models that explain the given training data well.
- Models that minimize the training error.

$$\underbrace{MSE(training) = (Y - \bar{Y})^2}$$

Minimize this!

Expected MSE

$$\text{Expected } MSE(Y) = \mathbb{E}[(Y - \bar{Y})^2]$$

$$= \mathbb{E}[(Y - \mathbb{E}(Y) + \mathbb{E}(Y) - \bar{Y})^2]$$

$$= \mathbb{E}[(Y - \mathbb{E}[Y])^2 + 2(Y - \mathbb{E}[Y])(\mathbb{E}[Y] - \bar{Y}) + (\mathbb{E}[Y] - \bar{Y})^2]$$

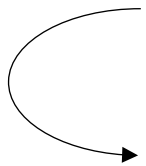
$$= \mathbb{E}[(Y - \mathbb{E}[Y])^2] + \mathbb{E}[2(Y - \mathbb{E}[Y])(\mathbb{E}[Y] - \bar{Y})] + \mathbb{E}[(\mathbb{E}[Y] - \bar{Y})^2]$$

$$= \mathbb{E}[(Y - \mathbb{E}[Y])^2] + 2\mathbb{E}[Y - \mathbb{E}[Y]] \cdot \mathbb{E}[\mathbb{E}[Y] - \bar{Y}] + \mathbb{E}[(\mathbb{E}(Y) - \bar{Y})^2]$$

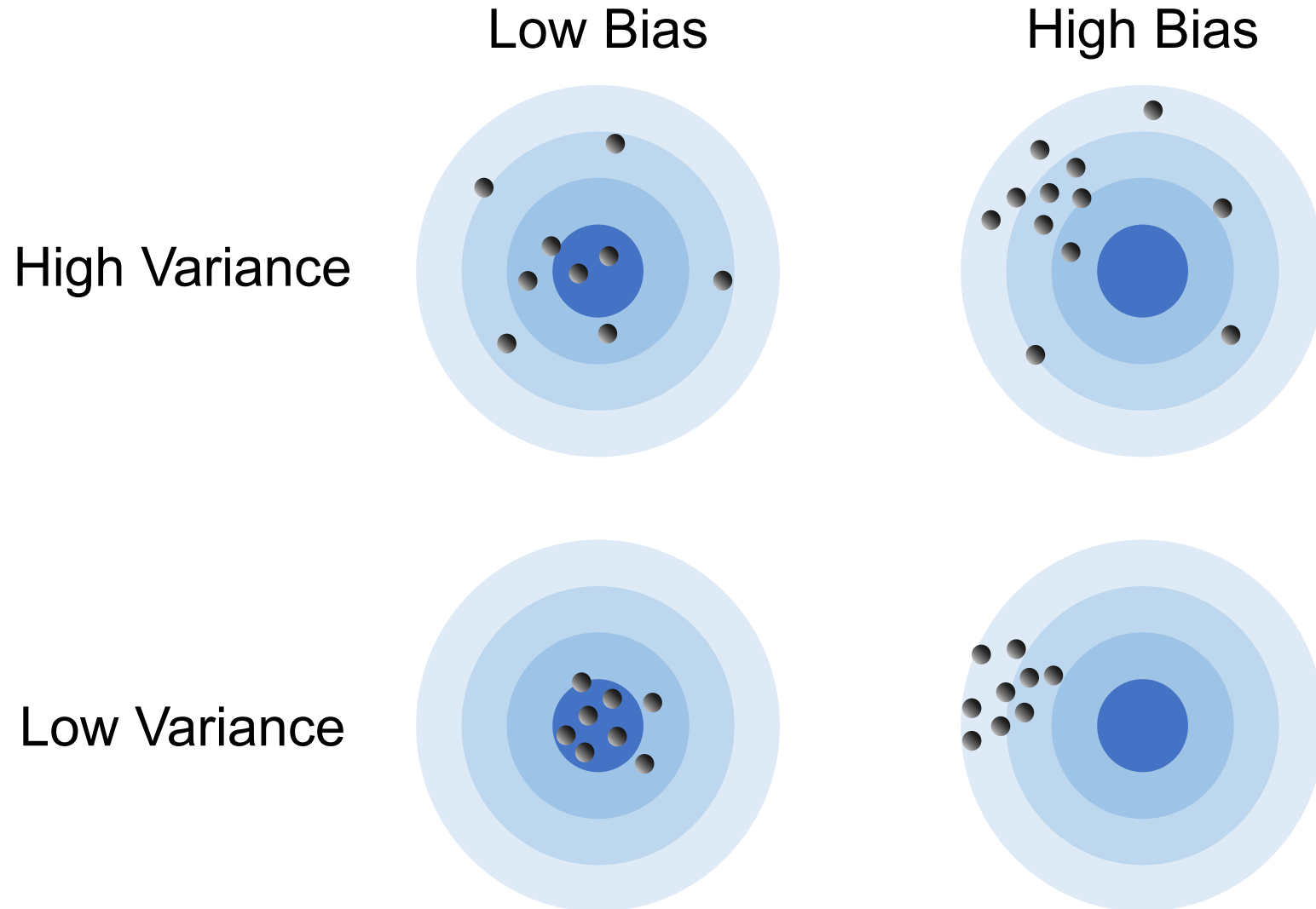
$$= \mathbb{E}[(Y - \mathbb{E}[Y])^2] + 2(\mathbb{E}[Y] - \mathbb{E}[Y]) \cdot \mathbb{E}[\mathbb{E}[Y] - \bar{Y}] + \mathbb{E}[(\mathbb{E}(Y) - \bar{Y})^2]$$

$$= \underbrace{\mathbb{E}[(Y - \mathbb{E}[Y])^2]}_{\text{Variance}} + \underbrace{\mathbb{E}[(\mathbb{E}(Y) - \bar{Y})^2]}_{\text{Bias}^2}$$

Law of iterated
expectation

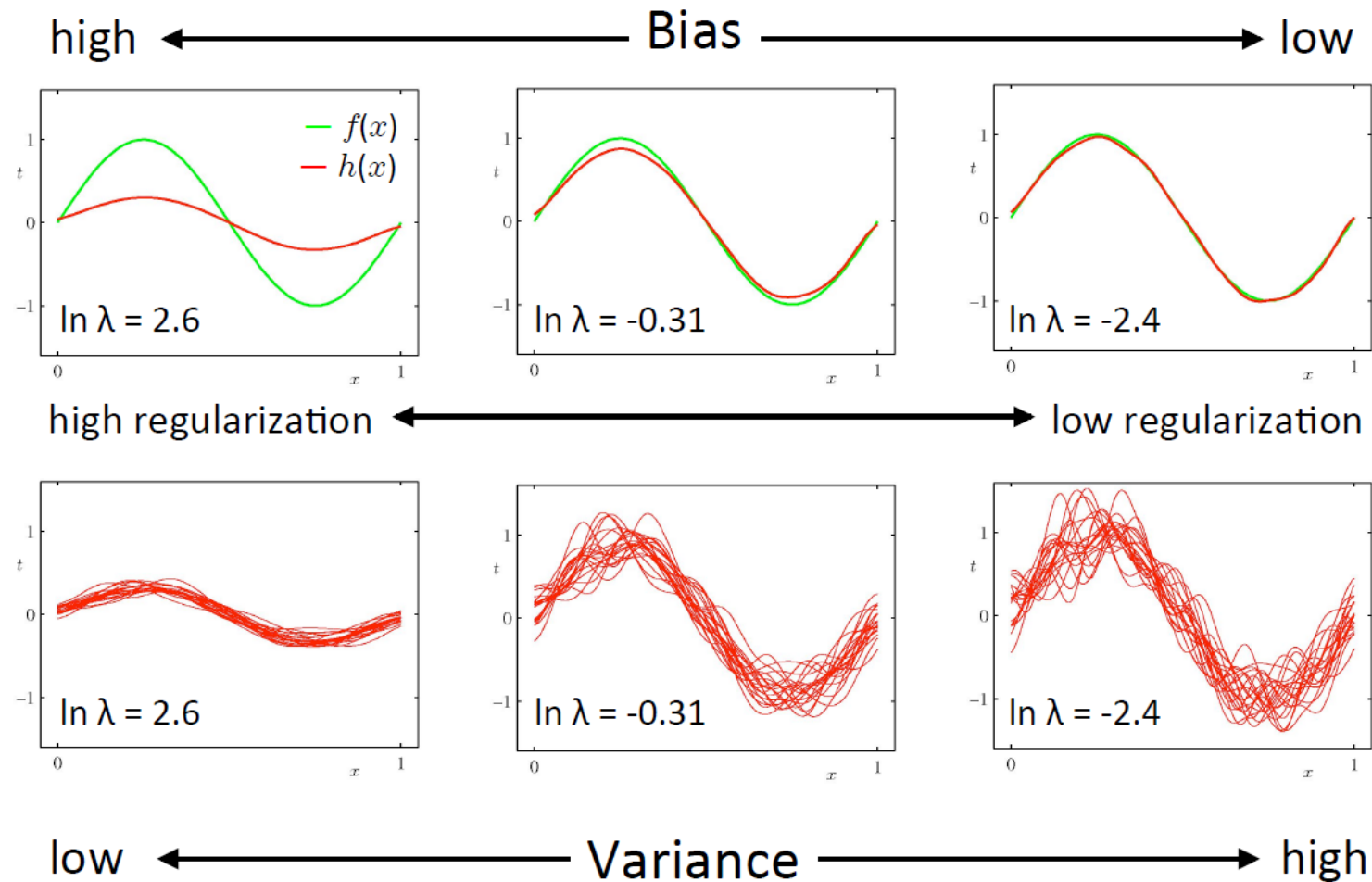


Variance vs. Bias



높은 분산, 낮은 편향
상태의 데이터를 낮은
분산, 높은 편향 상태
의 데이터로 유도하
는 것이 규제

Illustration of Bias-Variance



Good Predictive Model

- Models that well predict the output of unknown input data.
- Models that minimize the expected error on the test data.

$$\text{Expected } MSE(Y) = \underbrace{\mathbb{E}[(Y - \mathbb{E}[Y])^2]}_{\text{Variance}} + \underbrace{\mathbb{E}[(\mathbb{E}(Y) - \bar{Y})^2]}_{\text{Bias}^2}$$

- Training drives down bias, but ignores variance.
- ***To get a good predictive model, we have to care of variance!***

분산과 편향은 서로 트레이드 오프 관계이므로 분산 최소화를 위해 의도적으로 약간의 편향을 허용

Our Strategy: Reduce Variance

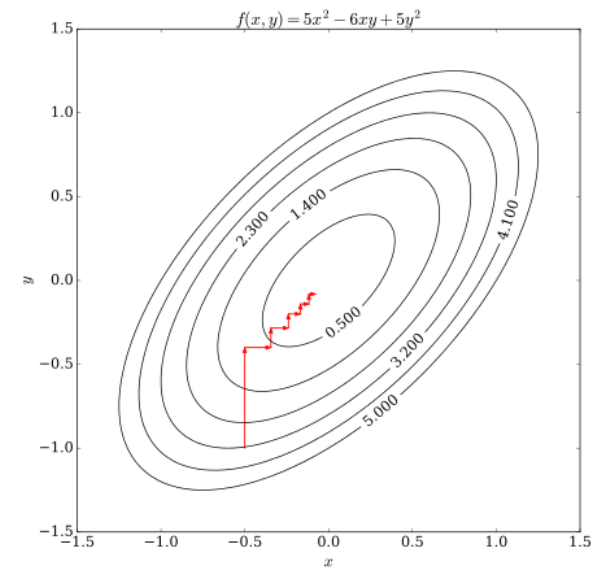
How do we reduce the variance at the cost of allowing bias?



Subset Selection Strategy

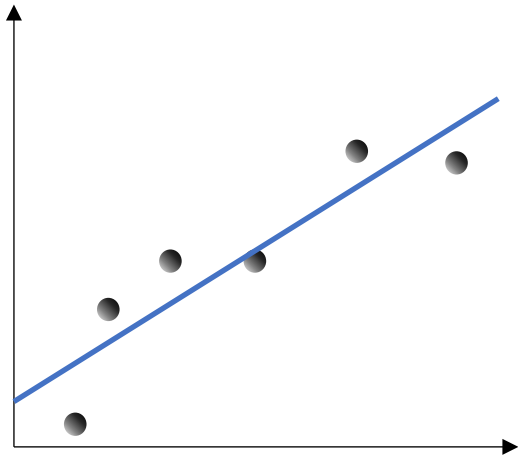
- Instead of training all the model parameters, selectively train a subset of them!
 - Increases the bias (we are training only a subset of the model!) while decreasing the variance.
 - Best subset selection
 - Forward stepwise selection
 - Backward stepwise selection
 - Least angle selection
 - Etc.

E.g., Stochastic Coordinate Descent (SCD)

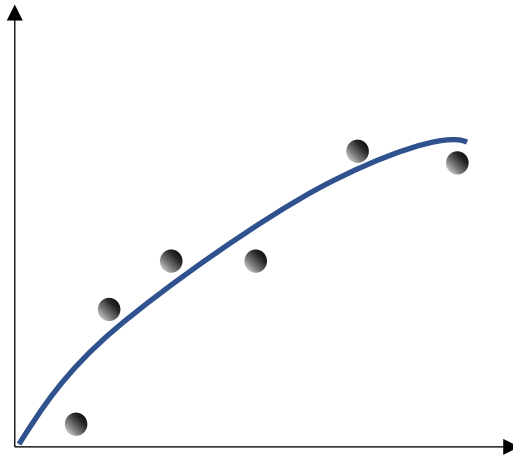


- Good Model?
- ***Regularization Concept***
- Ridge Regression (L_2 -Regularization)

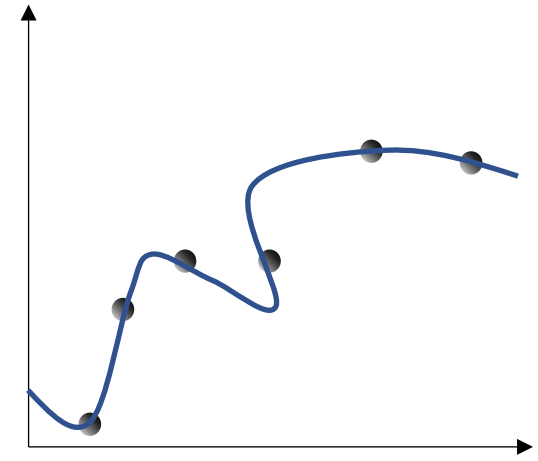
Regularization Concept (1/3)



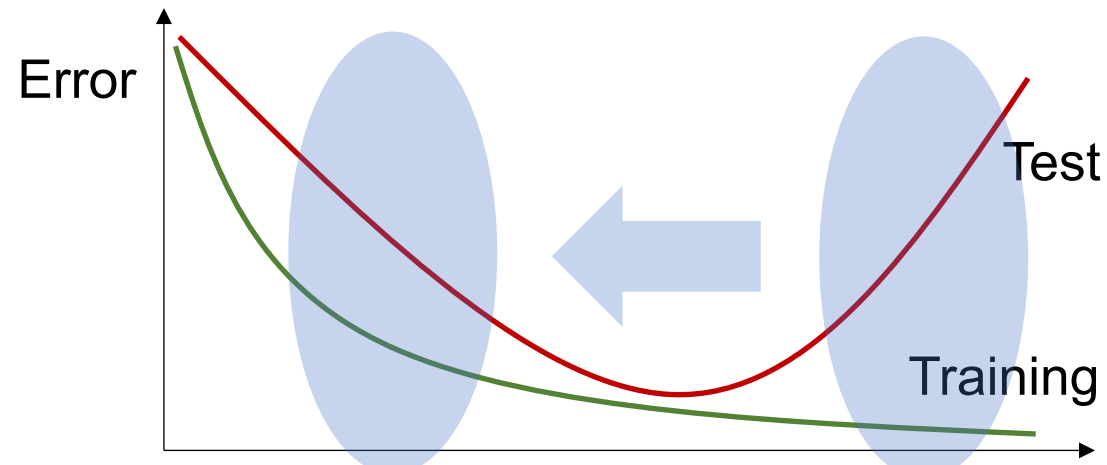
$$y = w_0 + w_1x$$



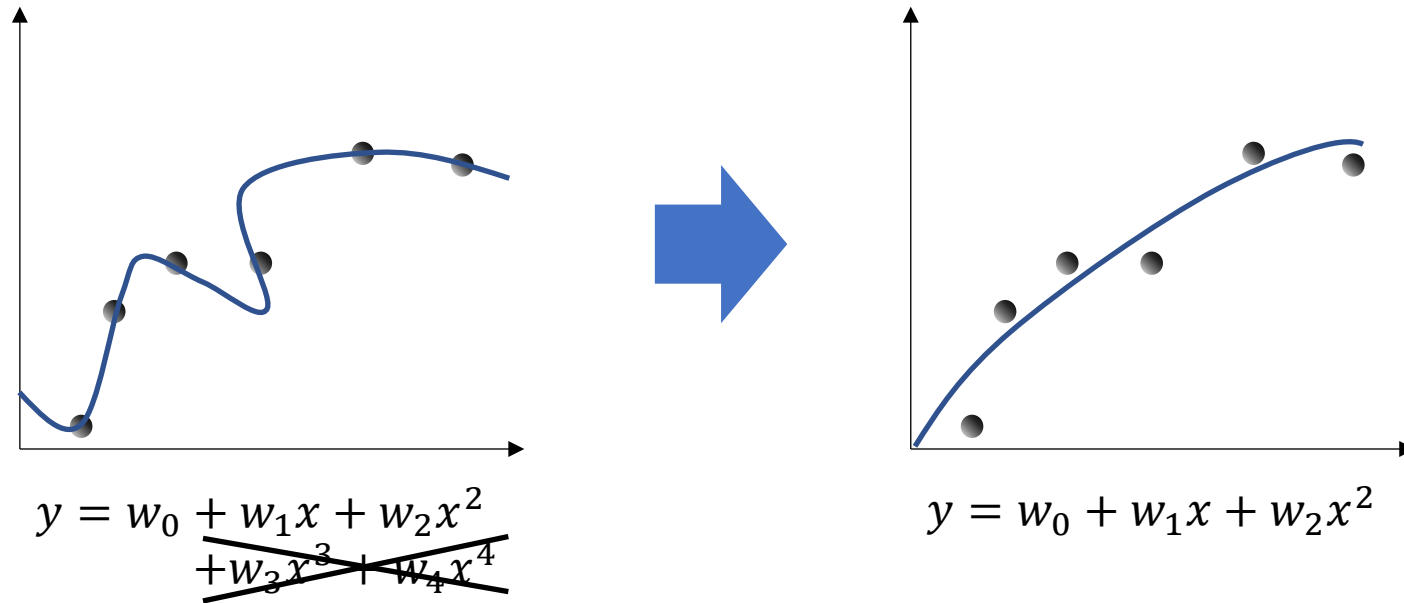
$$y = w_0 + w_1x + w_2x^2$$



$$y = w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4$$



Regularization Concept (2/3)



$$\min_w \sum_{i=1} (y - \bar{y})^2 + \underbrace{5000w_3^2 + 5000w_4^2}$$

The solution will likely have a very small w_3 and w_4 .

Regularization Concept (3/3)

$$w = w_0, w_1, \dots, w_d \quad y = w_0 + w_1x + \dots + w_dx^d$$

$$L(w) = \min_w \underbrace{\sum_{i=1} (y - \bar{y})^2}_{(1) \text{ Training accuracy}} + \lambda \underbrace{\sum_{j=1}^d w_j^2}_{(2) \text{ Generalization accuracy}}$$

Notice that we don't add w_0

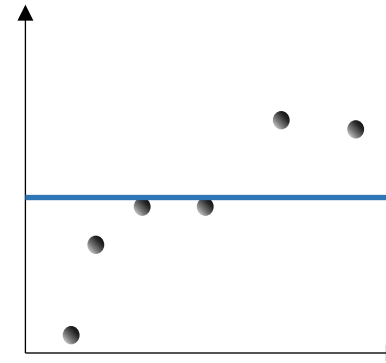
λ : the regularization parameter that controls the trade-off between (1) and (2).

The Regularization Parameter

SVM에서와는 다르게 고차항의 영향 최소화 자체를 문제로 정의하지는 않으며 고차항의 영향 억제를 하이퍼파라미터를 통해 어느정도 유도만 함

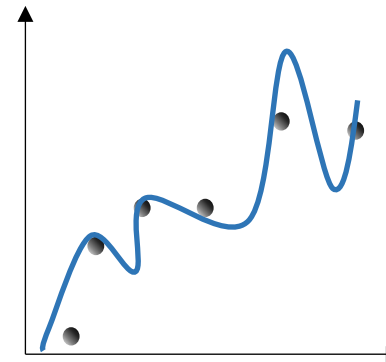
$$L(w) = \min_w \sum_{i=1} (y_i - \bar{y})^2 + \lambda \sum_{j=1}^d w_j^2$$

Very big $\lambda \rightarrow w_1 \approx 0, \dots, w_d \approx 0$



$y = w_0$
Underfitting!

Very small $\lambda \rightarrow$ very large w



$y = w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4$
Overfitting!

Regularization Method

- Regularization methods usually enforce restrictions on the magnitude of the regression parameters.
- The **variance** is expected to **decrease** while **increasing** the **bias**.

Example of Least Squares Method

$$L(w) = \min_{w_1, w_2} \sum_{i=1} (x_{i,1}w_1 + x_{i,2}w_2 - \bar{y})^2$$

Example of Regularization Method

$$L(w) = \min_{w_1, w_2} \sum_{i=1} (x_{i,1}w_1 + x_{i,2}w_2 - \bar{y})^2$$

subject to $w_1^2 + w_2^2 \leq 30$

Example: Which row is the solution?

$$L(w) = \min_{w_1, w_2} \sum_{i=1} (x_{i,1}w_1 + x_{i,2}w_2 - \bar{y})^2 + \lambda \sum_{j=1}^d w_j^2$$

subject to $w_1^2 + w_2^2 \leq 30$

(w_1, w_2)	$w_1^2 + w_2^2$	MSE
(4,5)	41	20
(3,5)	34	23
(4,4)	32	25
(2,5)	27	27
(2,4)	18	25
(2,3)	13	29

- Good Model?
- Regularization Concept
- ***Ridge Regression (L_2 -Regularization)***
- Lasso Regression (L_1 -Regularization)

Ridge Regression

- Linear regression with L_2 -norm regularization.
 - Minimize the loss (e.g., MSE) while penalizing the L_2 -norm of the model parameters.

$$L(w) = \underset{w}{\operatorname{argmin}} \sum_{i=1} (\mathbf{w}^T \mathbf{x}_i - \bar{y}_i)^2$$

subject to $\sum_{j=1}^d w_j^2 \leq t$



Equivalent (Lagrangian Multiplier)

$$L(w) = \underset{w}{\operatorname{argmin}} \left\{ \sum_{i=1} (\mathbf{w}^T \mathbf{x}_i - \bar{y}_i)^2 + \lambda \sum_{j=1}^d w_j^2 \right\}$$

$$\lambda \geq 0$$

MSE Contour (1/2)

- Let us consider a two-dimensional case.

$$\begin{aligned}MSE(w_1, w_2) &= \sum_{i=1}^n (y_i - w_1 x_{i,1} - w_2 x_{i,2})^2 \\&= \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n y_i (w_1 x_{i,1} + w_2 x_{i,2}) + \sum_{i=1}^n (w_1 x_{i,1} + w_2 x_{i,2})^2 \\&= \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n (y_i x_{i,1}) w_1 - 2 \sum_{i=1}^n (y_i x_{i,2}) w_2 + \sum_{i=1}^n w_1^2 x_{i,1}^2 + w_2^2 x_{i,2}^2 + 2w_1 w_2 x_{i,1} x_{i,2} \\&= \left(\sum_{i=1}^n x_{i,1}^2 \right) w_1^2 + \left(\sum_{i=1}^n x_{i,2}^2 \right) w_2^2 + \left(2 \sum_{i=1}^n x_{i,1} x_{i,2} \right) w_1 w_2 \\&\quad - 2 \left(\sum_{i=1}^n y_i x_{i,1} \right) w_1 - 2 \left(\sum_{i=1}^n y_i x_{i,2} \right) w_2 + \sum_{i=1}^n y_i^2 \\&= Aw_1^2 + Bw_1 w_2 + Cw_2^2 + Dw_1 + Ew_2 + F \quad \text{Conic equation (2-dimensional)}\end{aligned}$$

MSE Contour (2/2)

$$Aw_1^2 + Bw_1w_2 + Cw_2^2 + Dw_1 + Ew_2 + F = 0$$

Discriminant of conic equation (판별식): $B^2 - 4AC$

$$B^2 - 4AC = 0 \rightarrow \text{parabola (포물선)}$$

$$B^2 - 4AC > 0 \rightarrow \text{hyperbola (쌍곡선)}$$

$$B^2 - 4AC < 0 \rightarrow \text{ellipse (타원)}$$

$$B = 0 \text{ and } A = C \rightarrow \text{circle (원)}$$

$$\begin{aligned} = & \left(\sum_{i=1}^n x_{i,1}^2 \right) w_1^2 + \left(\sum_{i=1}^n x_{i,2}^2 \right) w_2^2 + \left(2 \sum_{i=1}^n x_{i,1} x_{i,2} \right) w_1 w_2 \\ & - 2 \left(\sum_{i=1}^n y_i x_{i,1} \right) w_1 - 2 \left(\sum_{i=1}^n y_i x_{i,2} \right) w_2 + \sum_{i=1}^n y_i^2 \end{aligned}$$

$$\begin{aligned} B^2 - 4AC &= \left(2 \sum_{i=1}^n x_{i,1} x_{i,2} \right)^2 - 4 \sum_{i=1}^n x_{i,1}^2 \sum_{i=1}^n x_{i,2}^2 \\ &= 4 \left\{ \left(\sum_{i=1}^n x_{i,1} x_{i,2} \right)^2 - \sum_{i=1}^n x_{i,1}^2 \sum_{i=1}^n x_{i,2}^2 \right\} < 0 \end{aligned}$$

By Cauchy-Schwarz inequality
 $(\langle a, b \rangle)^2 \leq \langle a, a \rangle \langle b, b \rangle$

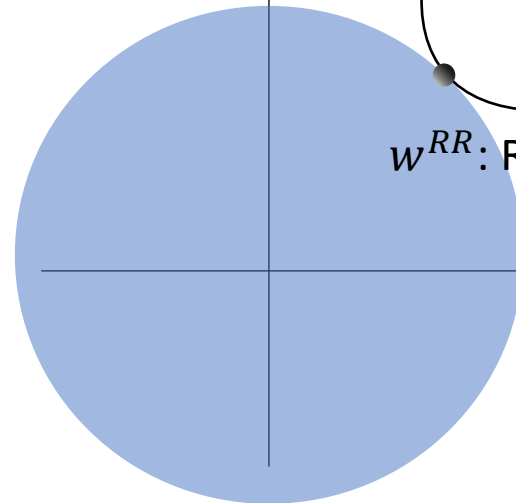
Ridge Regression Schematic Illustration

$$L(w) = \underset{w}{\operatorname{argmin}} \sum_{i=1}^n (w^T x_i - \bar{y}_i)^2$$

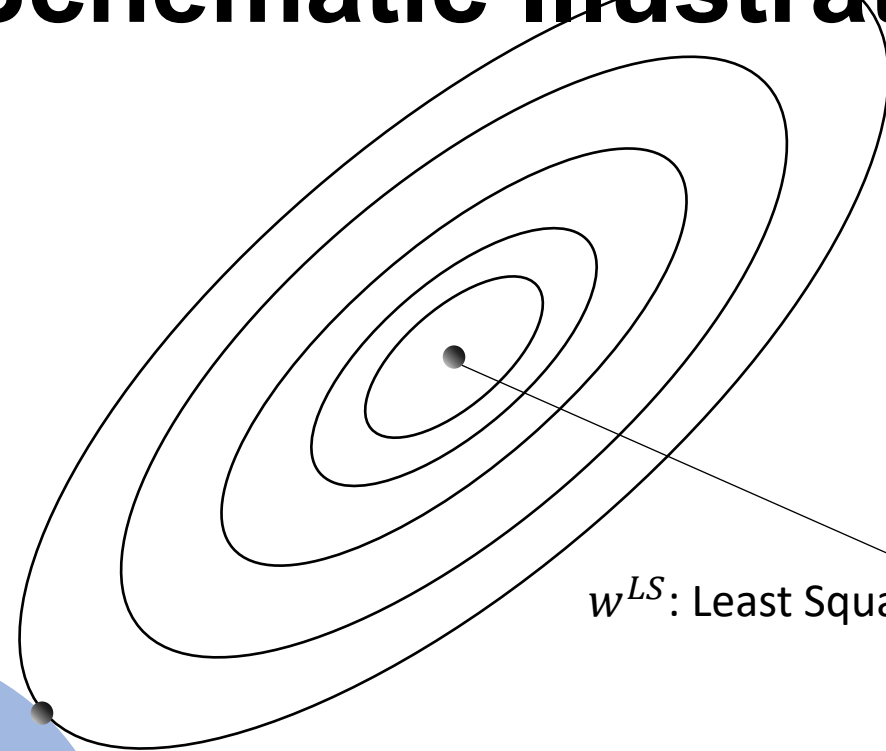
subject to $\sum_{j=1}^d w_j^2 \leq t$

Remember that $B^2 - 4AC \leq 0 \rightarrow$ **Ellipse**

$$w_1^2 + w_2^2 \leq t$$

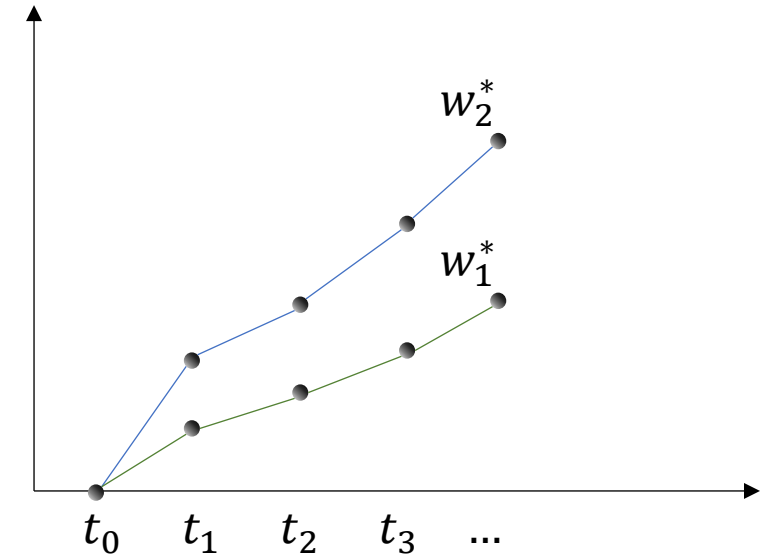
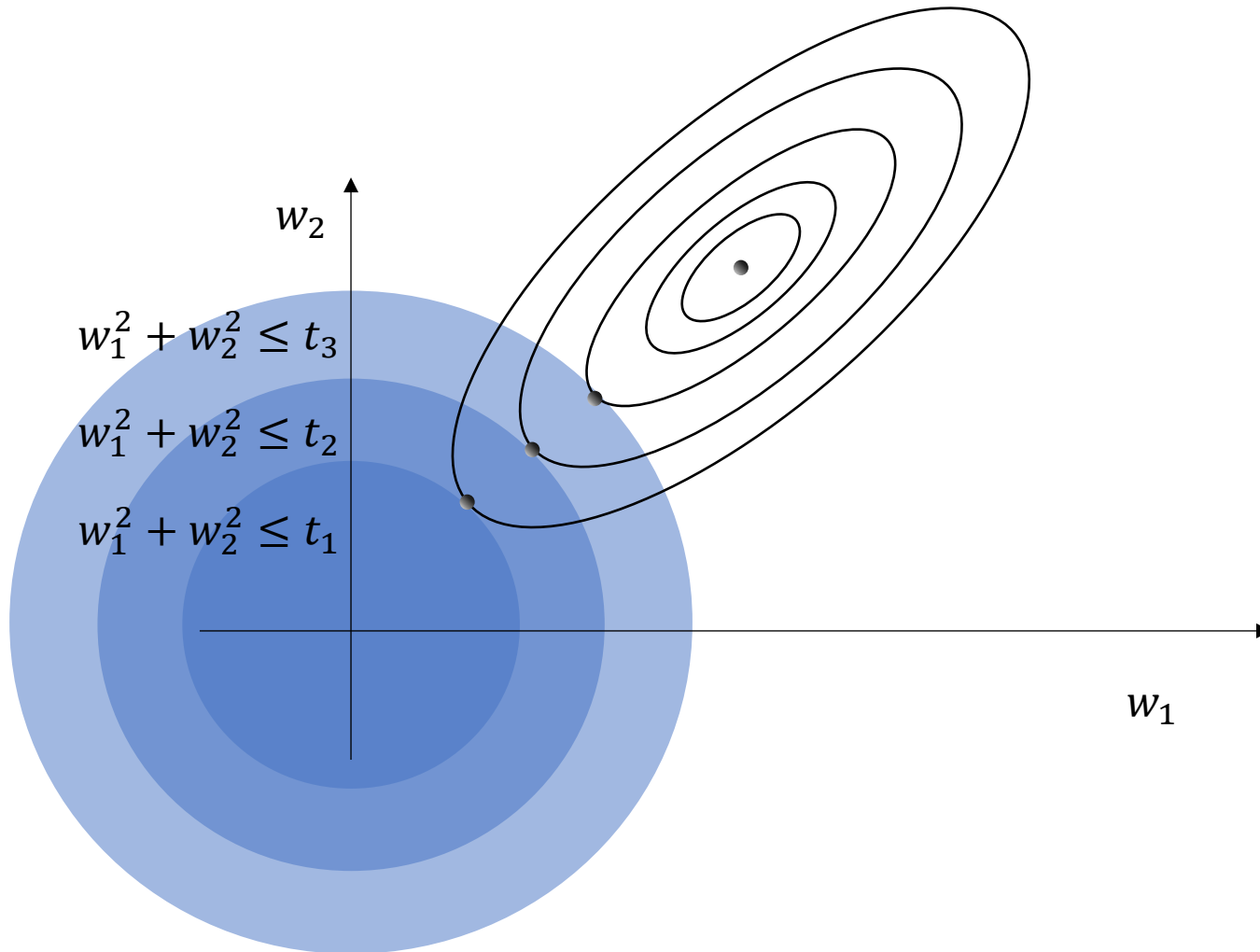


w^{RR} : Ridge Regression estimator



w^{LS} : Least Square estimator

Ridge Solution Path



$$L(w) = \operatorname{argmin}_w \sum_{i=1}^n (w^T x_i - \bar{y}_i)^2$$

subject to $\sum_{j=1}^d w_j^2 \leq t$

Closed Form Solution

- The closed form solution of Ridge Regression:

$$\begin{aligned} L(w) &= (y - Xw)^T (y - Xw) + \lambda w^T w \\ &= y^T y + w^T X^T X w - 2w^T X^T y + w^T \lambda I_p w \end{aligned}$$

$$\frac{\partial L(w)}{\partial w} = 2X^T X w - 2X^T y + 2\lambda I_p w = 0$$

$$(X^T X + \lambda I_p)w - X^T y = 0$$

$$(X^T X + \lambda I_p)w = X^T y$$

$$\mathbb{E}[w] = (X^T X + \lambda I_p)^{-1} X^T y$$

Remember that the solution of Linear regression was

$$w^* = (X^T X)^{-1} X^T y$$

Bias of the Estimator w

$$\mathbb{E}[w] = (X^T X + \lambda I_p)^{-1} X^T y$$

람다를 크게 하면 자연스럽게 원래 해답보다 편향이 커짐

- Bias = $\mathbb{E}[w] - w$

$$= (X^T X + \lambda I_p)^{-1} X^T y - w$$

$$= (X^T X + \lambda I_p)^{-1} X^T X w - w$$

$$= (X^T X + \lambda I_p)^{-1} X^T X w - (X^T X)^{-1} X^T X w$$

$$= \left((X^T X + \lambda I_p)^{-1} - (X^T X)^{-1} \right) X^T X w$$

Observe that as $\lambda \rightarrow 0$, the bias goes to 0.

Variance of the Estimator w

- Variance = $\mathbb{E}[\mathbb{E}[w] - w]$

We skip the proof of this.. Consult [1].

$$= \sigma^2 (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1}$$

Observe that as $\lambda \rightarrow \infty$, the variance goes to 0.

$$((cA)^{-1} = c^{-1}A^{-1})$$

편향과 분산의 트레이드 오프 관계가 나타남

Impact of Regularization

- As having a certain non-zero λ , we reduce the variance while increasing the bias.
- This implies that the *expected MSE* decreases improving the *generalization*.

- Good Model?
- Regularization Concept
- Ridge Regression (L_2 -Regularization)
- ***Lasso Regression (L_1 -Regularization)***

Lasso Regression

- **L**east **A**bsolute **S**hrinkage and **S**election **O**perator (LASSO)
 - Minimize the loss (e.g., MSE) while penalizing the L_1 -norm of the model parameters.

$$L(w) = \underset{w}{\operatorname{argmin}} \sum_{i=1} (\mathbf{w}^T \mathbf{x}_i - \bar{y}_i)^2$$

subject to $\sum_{j=1}^d |w_j| \leq t$



Equivalent (Lagrangian Multiplier)

$$L(w) = \underset{w}{\operatorname{argmin}} \left\{ \sum_{i=1} (\mathbf{w}^T \mathbf{x}_i - \bar{y}_i)^2 + \lambda \sum_{j=1}^d |w_j| \right\}$$

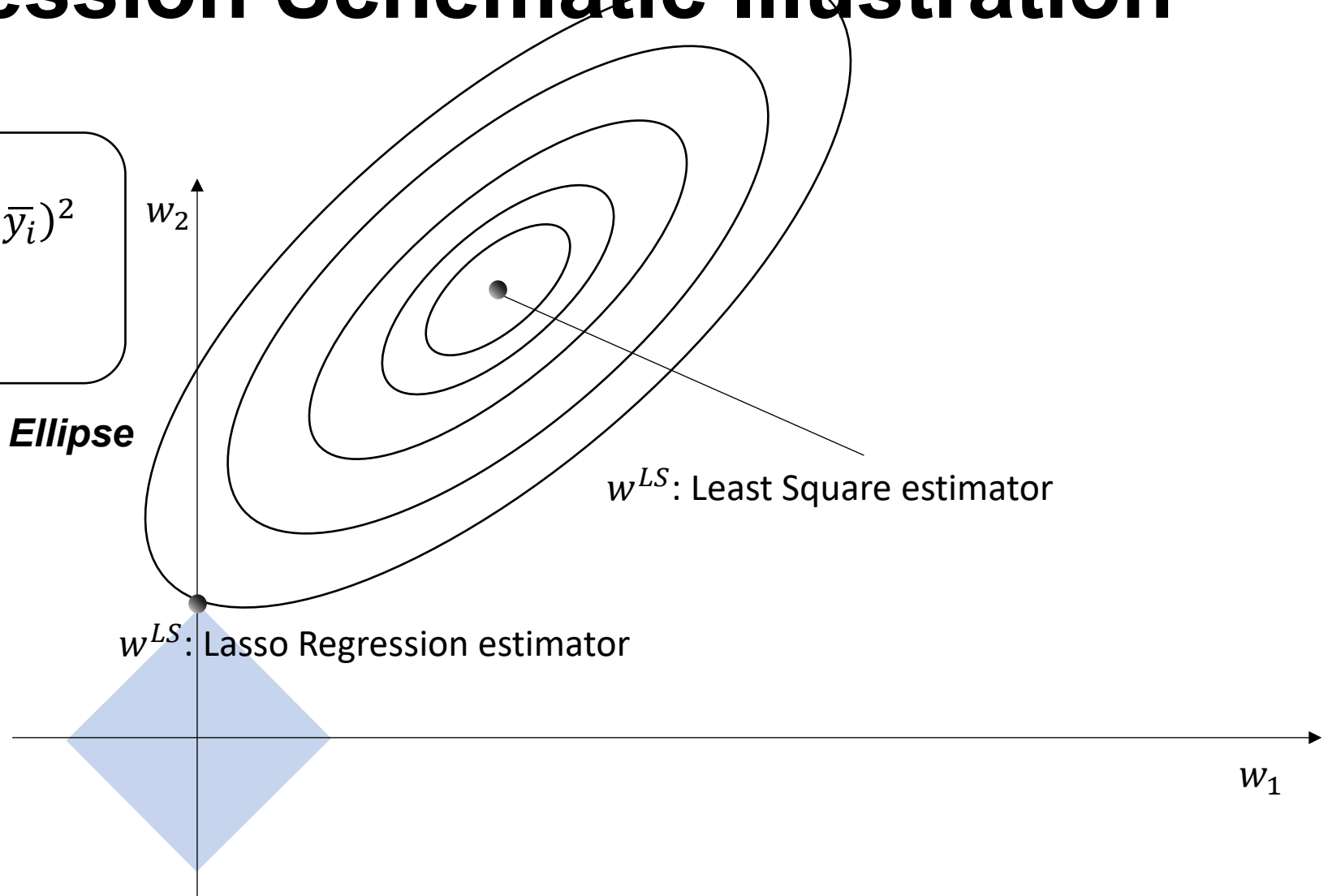
Lasso Regression Schematic Illustration

$$L(w) = \underset{w}{\operatorname{argmin}} \sum_{i=1}^n (w^T x_i - \bar{y}_i)^2$$

subject to $\sum_{j=1}^d |w_j| \leq t$

Remember that $B^2 - 4AC \leq 0 \rightarrow$ **Ellipse**

$$|w_{j,1}| + |w_{j,2}| \leq t$$



Lasso Solution

Ridge Solution

$$L(w) = \operatorname{argmin}_w \left\{ \sum_{i=1} (\mathbf{w}^T \mathbf{x}_i - \bar{y}_i)^2 + \lambda \sum_{j=1}^d w_j^2 \right\}$$

$$\Rightarrow w^* = (X^T X + \lambda I_p)^{-1} X^T y$$

Lasso Solution

$$L(w) = \operatorname{argmin}_w \left\{ \sum_{i=1} (\mathbf{w}^T \mathbf{x}_i - \bar{y}_i)^2 + \lambda \sum_{j=1}^d |w_j| \right\}$$

$$\Rightarrow w^* = ??$$

- Lasso does not have a closed form solution... why?
 - Because L_1 -norm is not differentiable.
- Instead of the closed form solution, usually we use quadratic programming techniques to solve the Lasso.

Parameter Selection with Lasso

With a sufficiently small t , a few parameters are supposed to be zero.

This effect can be used to select important parameters!

$$|w_{j,1}| + |w_{j,2}| \leq t'$$

$$|w_{j,1}| + |w_{j,2}| \leq t$$

With a large t' , the solution may have no zero parameters... → limited regularization effect.

Any Questions?

Regression is used in most of ML techniques!

